

Probability Theory - Winter term 2025/26

Personal Solution Set 2

TU Dortmund MS Data Science

Problem 1

The Borel σ -field $\mathcal{B}(\mathbb{R})$ on $\mathbb{R}$ can be defined equivalently in different ways. Here, we define $\mathcal{B}(\mathbb{R})$ as the smallest σ -field on $\mathbb{R}$ that contains all half-open intervals, $(a, b]$, $-\infty < a < b < \infty$, as elements. Show that $\mathcal{B}(\mathbb{R})$ contains the following sets:

1. one point sets $\{x\}$, where $x \in \mathbb{R}$,
2. open intervals (a, b) , $-\infty < a < b < \infty$
3. intervals $(-\infty, b)$ and $(-\infty, b]$, $b \in \mathbb{R}$

Solution

1. Let $x \in \mathbb{R}$. Define the sequence $A_n := (x - \frac{1}{n}, x]$ for $n \in \mathbb{N}$. For all n , $A_n \in \mathcal{B}(\mathbb{R})$. Since σ -fields are closed under countable intersections, I have:

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x\right] = \{x\} \in \mathcal{B}(\mathbb{R}).$$

2. Let $a < b$. By definition, $(a, b] \in \mathcal{B}(\mathbb{R})$. I can write $(a, b) = (a, b] \setminus \{b\}$. Using the result from item 1, $\{b\} \in \mathcal{B}(\mathbb{R})$. Since σ -fields are closed under set differences, $(a, b) \in \mathcal{B}(\mathbb{R})$.
3. Let $b \in \mathbb{R}$. Define the sequence $A_n := (-n, b]$ for $n \in \mathbb{N}$. By definition, $A_n \in \mathcal{B}(\mathbb{R})$. Since σ -fields are closed under countable unions, I get:

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} (-n, b] = (-\infty, b] \in \mathcal{B}(\mathbb{R}).$$

For the open interval, I write $(-\infty, b) = (-\infty, b] \setminus \{b\}$. By previous results, both sets are in $\mathcal{B}(\mathbb{R})$, so $(-\infty, b) \in \mathcal{B}(\mathbb{R})$.

Problem 2

Let $\Omega = \mathbb{N}$ and define the family of subsets $\mathcal{A}$ of Ω by

$$\mathcal{A} = \{A \subset \Omega \text{ with } |A| < \infty \vee |A^c| < \infty\}.$$

- a) Show that $\mathcal{A}$ is a field. Is it also a σ -field?
- b) What is $\sigma(\mathcal{A})$, i.e. the σ -field generated by $\mathcal{A}$?
- c) Define $\mu : \mathcal{A} \rightarrow [0, \infty]$ by $\mu(A) = |A|$ (the counting measure). Is μ a pre-measure on $\mathcal{A}$?

Solution

a) Using definition 1.13, I check the three properties of a field:

- (i) $|\emptyset| = 0 < \infty$, so $\emptyset \in \mathcal{A}$.
- (ii) Let $A \in \mathcal{A}$. If $|A| < \infty$, then its complement A^c has $|(A^c)^c| = |A| < \infty$, so $A^c \in \mathcal{A}$. If $|A^c| < \infty$, then $A^c \in \mathcal{A}$ immediately.
- (iii) Let $A_1, A_2 \in \mathcal{A}$. If $|A_1| < \infty$ and $|A_2| < \infty$, then $|A_1 \cup A_2| < \infty$, so $A_1 \cup A_2 \in \mathcal{A}$. If $|A_1^c| < \infty$, then $(A_1 \cup A_2)^c = A_1^c \cap A_2^c \subseteq A_1^c$. Thus $|(A_1 \cup A_2)^c| < \infty$, so $A_1 \cup A_2 \in \mathcal{A}$. The same holds if $|A_2^c| < \infty$.

$\mathcal{A}$ is a field. However, it is not a σ -field. Let $A_i = \{2i\}$ for $i \in \mathbb{N}$. Each $|A_i| = 1 < \infty$, so $A_i \in \mathcal{A}$. But $\bigcup_{i=1}^{\infty} A_i$ is the set of even numbers, which is infinite, and its complement is the set of odd numbers, which is also infinite. Thus, $\bigcup_{i=1}^{\infty} A_i \notin \mathcal{A}$.

b) Since $\mathcal{A}$ contains all singletons of $\mathbb{N}$ (as they are finite), $\sigma(\mathcal{A})$ must contain all countable unions of singletons. Since $\mathbb{N}$ is countable, this generates all subsets of $\mathbb{N}$. Therefore, $\sigma(\mathcal{A}) = \mathcal{P}(\mathbb{N})$.

c) Using definition 1.13, μ is a pre-measure if it is a positive mapping and σ -additive for disjoint sequences whose union is in $\mathcal{A}$. $\mu(A) = |A| \geq 0$, so it is a positive mapping. Let $A_1, A_2, \dots$ be pairwise disjoint sets in $\mathcal{A}$ such that $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$. If all A_i are finite, $\mu(\bigcup_{i=1}^{\infty} A_i) = |\bigcup_{i=1}^{\infty} A_i| = \sum_{i=1}^{\infty} |A_i| = \sum_{i=1}^{\infty} \mu(A_i)$. If at least one A_k is infinite, then $\sum_{i=1}^{\infty} \mu(A_i) \geq \mu(A_k) = \infty$, and $\mu(\bigcup_{i=1}^{\infty} A_i) = \infty$. Thus, μ is a pre-measure.

Problem 3

Let $(\Omega, \mathcal{A})$ be a measurable space and let ν_1 and ν_2 be finite measures on $(\Omega, \mathcal{A})$. Prove or disprove that μ_i for $i = 1, 2, 3$ is a measure.

1. $\mu_1 = \nu_1 \cdot \nu_2$
2. $\mu_2 = \min\{\nu_1, \nu_2\}$
3. $\mu_3 = \nu_2 - \nu_1$

Solution

Let $\Omega = \mathbb{N}$, $\mathcal{A} = \mathcal{P}(\mathbb{N})$, and choose Dirac measures $\nu_1 = \delta_{\{1\}}$ and $\nu_2 = \delta_{\{2\}}$. Let $A = \{1\}$ and $B = \{2\}$. A and B are disjoint.

1. μ_1 is not a measure. It fails σ -additivity:

$$\mu_1(A \cup B) = \nu_1(\{1, 2\}) \cdot \nu_2(\{1, 2\}) = 1 \cdot 1 = 1.$$

$$\mu_1(A) + \mu_1(B) = (\nu_1(\{1\}) \cdot \nu_2(\{1\})) + (\nu_1(\{2\}) \cdot \nu_2(\{2\})) = (1 \cdot 0) + (0 \cdot 1) = 0.$$

Since $1 \neq 0$, σ -additivity does not hold.

2. μ_2 is not a measure. It fails σ -additivity:

$$\mu_2(A \cup B) = \min\{\nu_1(\{1, 2\}), \nu_2(\{1, 2\})\} = \min\{1, 1\} = 1.$$

$$\mu_2(A) + \mu_2(B) = \min\{\nu_1(\{1\}), \nu_2(\{1\})\} + \min\{\nu_1(\{2\}), \nu_2(\{2\})\} = 0 + 0 = 0.$$

Since $1 \neq 0$, σ -additivity does not hold.

3. μ_3 is not a measure. It fails the positive mapping requirement:

$$\mu_3(A) = \nu_2(\{1\}) - \nu_1(\{1\}) = 0 - 1 = -1 < 0.$$